ANXIETYSENSE: A MULTI MODEL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

25-26J-376

TEAM MEMBERS

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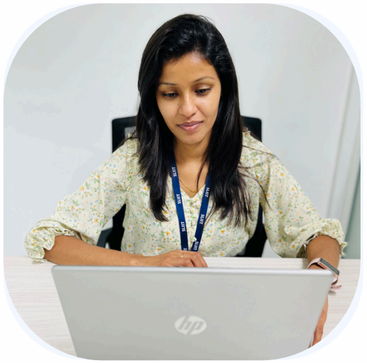

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PROBLEM STATEMENT & STAKEHOLDERS

Context

- Domain: Digital Mental Health Solutions
- Setting: Sri Lanka (Resource-constrained healthcare environment)
- Organization: Schools, Hospitals, Community Centers, Homes

STAKEHOLDERS

Primary Users	Secondary Users	Decision Makers
Children (8 -14 yrs)	Parents & Teachers	Ministry of Helath
Young Adult (18 - 35yrs)	Family Members	Ministry of Education
Pregnet Women	Healthhcare Providers	Hospital Administrators
Eldery (60+ yrrs)	School Counselors	Insuarance Providers

Observed Issue

- 1 in 7 adolescents experience mental health disorders (WHO 2023)
- 20% of pregnant women experience perinatal anxiety
- 15-20% of elderly suffer from clinically significant anxiety
- >20% of young adults experience clinically significant anxiety
- 70% of Sri Lankan schools lack minimum counseling facilities

Why It Matters

- Untreated anxiety leads to increased healthcare costs
- Lower productivity and academic performance
- Higher dropout rates and social isolation
- Long-term mental health complications in adulthood

RESEARCH GAP

Existing Solutions	Our Solution
Single-modality assessment (questionnaire OR AI)	Multimodal AI (Questionnaire + Facial + Voice)
Cloud-based processing with privacy concerns	On-device processing for privacy protection
Generic interventions without life-stage specificity	Life-stage-specific anxiety models
No explainable AI for user trust	User Friendly Chatbot

OBJECTIVE

To design and develop a comprehensive mobile application system that improves anxiety detection and management across all life stages through the integration of validated psychological assessment tools with AI-driven emotion recognition and culturally adapted therapeutic interventions.

SOLUTION OVERVIEW

AnxietySense is a multimodal AI ecosystem that combines real-time facial emotion recognition and voice analysis with validated psychological questionnaires to provide life-stage-specific anxiety detection and personalized interventions for Sri Lankan populations.

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AI-Based Child Anxiety Prediction Using Vision and Behavioral Data

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1. TRADITIONAL ANXIETY ASSESSMENTS RELY ON QUESTIONNAIRES, WHICH MAY BE INACCURATE FOR CHILDREN.
2. SINGLE-METHOD SYSTEMS PROVIDE LIMITED ACCURACY.
3. THIS SYSTEM COMBINES QUESTIONNAIRES AND FACIAL EMOTION DETECTION.
4. CHILDREN AND PARENTS ANSWER A STANDARD QUESTIONNAIRE.
5. FACIAL EMOTIONS ARE DETECTED USING A MOBILENET MODEL.
6. BOTH RESULTS ARE COMBINED TO IMPROVE ANXIETY PREDICTION ACCURACY.

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MULTIMODAL YOUNG ADULT ANXIETY MANAGEMENT ANALYSIS

1. ANXIETY AND STRESS AMONG YOUNG ADULTS ARE INCREASING.
2. MOST ASSESSMENTS RELY ONLY ON QUESTIONNAIRES, WHICH CAN BE INACCURATE.
3. THIS SYSTEM USES AI TO IMPROVE MENTAL HEALTH ASSESSMENT.
4. THE DASS-21 QUESTIONNAIRE WITH A RANDOM FOREST MODEL PREDICTS ANXIETY LEVELS.
5. FACIAL EMOTIONS ARE DETECTED USING A CNN THROUGH THE CAMERA.
6. BOTH RESULTS ARE COMBINED TO PROVIDE THE FINAL ANXIETY LEVEL.

Current completion status

91%

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PREGNANCY – SPECIFIC ANXIETY DETECTION WITH VOICE ANALYSIS

1. QUESTIONNAIRES CAN MISS EMOTIONS, AND MANUAL ASSESSMENTS ARE SLOW.
2. THIS SYSTEM DETECTS EMOTIONS FROM VOICE USING AN LSTM MODEL.
3. INVALID RECORDINGS ARE REJECTED, AND INSTANT FEEDBACK IS GIVEN.
4. RESULTS ARE COMBINED WITH PASS QUESTIONNAIRE SCORES.
5. THE SYSTEM PROVIDES FAST, ACCURATE, AND SECURE ANXIETY PREDICTIONS.

Current completion status

91%

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AI-POWERED, CULTURALLY-ADAPTED MENTAL HEALTH SUPPORT FOR ELDERLY

OLD ADULTS - SPECIFIC ANXIETY DETECTION WITH PERSONALIZED EMOTIONALLY MUSIC THERAPY

1. ANXIETY IN ADULTS 60+ IS OFTEN UNDERDIAGNOSED.
2. TRADITIONAL ASSESSMENTS ARE SLOW AND CLINIC-DEPENDENT.
3. THIS SYSTEM USES GAS AND FACIAL EMOTION RECOGNITION FOR DETECTION.
4. PROVIDES CULTURALLY FAMILIAR MUSIC THERAPY FOR SUPPORT.
5. ELDER-FRIENDLY WITH LARGE TEXT, CLEAR BUTTONS, AND MINIMAL STEPS.

CONCLUSION & FUTURE IMPROVEMENTS

Conclusion

- AnxietySense provides AI-based anxiety detection for different life stages.
- Combines questionnaires, facial emotion recognition, and voice analysis.
- Helps support early detection and personalized mental health assistance.
- Current system development is over 90% completed.

Remaining Work / Improvements

- Final system testing and validation.
- UI improvements and usability enhancements for a better user experience.
- Optimization of AI model performance and accuracy.
- Final documentation and research paper completion.

THANK YOU

Feel free to share, support and help each other!